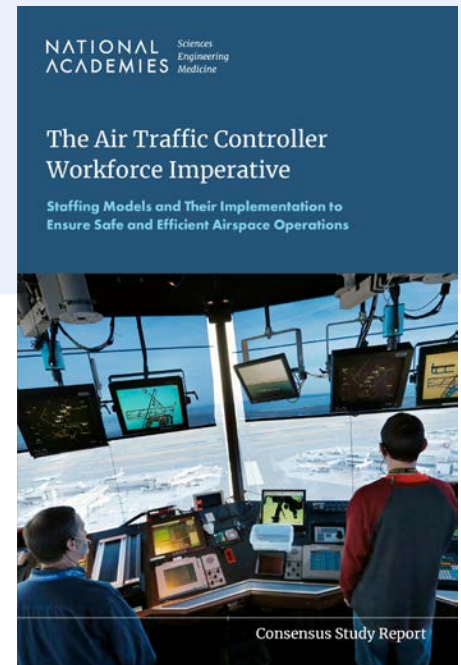


The Air Traffic Controller Workforce Imperative

Staffing Models and Their Implementation to Ensure Safe and Efficient Airspace Operations

The adequacy of controller staffing levels at the Federal Aviation Administration's (FAA's) 313 air traffic control (ATC) facilities is a matter of long-standing concern and has come under increased scrutiny. For decades, FAA has used staffing models for planning the controller workforce, which consisted of about 13,460 Full-Time Equivalents (FTEs) at the end of Fiscal Year (FY) 2024. About three quarters are fully Certified Professional Controllers (CPCs) who perform most of the work controlling traffic. FAA's staffing models and annual Controller Workforce Plans strive to provide staffing levels that are sized for safe and efficient air traffic operations at each facility. Safety is paramount. When ATC facilities are understaffed relative to the demands of traffic, FAA restricts the number of flights to levels the workforce can handle safely; however, implementing such measures can cause delays that cascade on busy traffic days.

At the request of Congress and commissioned by FAA,¹ an independent committee convened by the National Academies of Sciences, Engineering, and Medicine (NASEM) reviewed FAA's staffing models, their role in ensuring adequate staffing levels, and the factors contributing to staffing shortages at many of the agency's ATC facilities. The committee's report, *The Air Traffic Controller Workforce Imperative: Staffing Models and Their Implementation to Ensure Safe and Efficient Airspace Operations*, identifies past hiring constraints and other factors that continue to affect facility staffing levels. The report also compares FAA's traditional staffing models with an alternative approach and assesses FAA's response to



¹ FAA Reauthorization Act of 2024.

recommendations in an earlier NASEM report regarding controller fatigue and facility safety metrics.^{2,3}

KEY FINDINGS

Factors Affecting Current Staffing Levels

Between FYs 2013 and 2023, FAA experienced a series of constraints on hiring that were mostly externally imposed, including two government shutdowns and disruptions from the COVID-19 pandemic. These constraints led to hiring levels that did not keep pace with attrition, leading to reduced staffing at many facilities, including several facilities that serve the country's busiest airports and have an outsized effect on flight delays across the national airspace. Overall, FAA's controller workforce declined by almost 2,000 FTEs (–13%) between 2010 and 2024. Nineteen of the largest ATC facilities have controller staffing levels that are more than 15% below the staffing targets determined by FAA staffing models. While representing only 6% of facilities, these large, understaffed facilities serve about 27% of the country's commercial operations and account for about 40% of system delays.

Insufficient hiring has been a serious problem, but it has not been the only cause of understaffed facilities. FAA has not been successful in balancing the controller workforce across facilities; for instance, by inducing transfers from overstaffed to understaffed facilities. As a result, staffing levels at about 30% of facilities are more than 10% *below* their staffing targets, while the staffing levels at another 30% of facilities are more than 10% *above* their targets.

Overtime use at facilities has grown from 2% to 9% per year since hiring was constrained beginning in FY 2013. The increase in overtime has not been limited to facilities that are understaffed. Growth in overtime appears to be a widespread trend, perhaps due to inefficient shift scheduling of available controllers at facilities. Despite having procured a shift scheduling software package more than a decade ago, FAA did not implement it for facility use after testing in collaboration with the

National Air Traffic Controllers Association (NATCA). As a result, the agency's ATC facilities lack a robust shift scheduling tool to ensure that schedules comply with fatigue rules and policies while making the most efficient use of available staff. Other perplexing trends observed since FY 2010 that may be related to inefficient shift scheduling include a 4% reduction in regular hours worked per FTE and a 4% reduction in time on position (TOP) during regular hours worked per FTE. These trends are difficult to square with the 4% increase in operations and 13% decline in total FTEs during the period.

With support from Congress in FY 2024, FAA has been increasing its hiring of controllers and expanding its training capacity. A significant issue has been the shortage of former FAA controllers willing to serve as classroom trainers at the Academy and at individual facilities. While FAA has addressed this shortage at the Academy, it has struggled to reduce trainer shortages at facilities, particularly in high-cost-of-living areas where few former controllers continue to reside.

Expanded and improved training is needed to remedy declining training success rates and decrease the time required for new hires to fully certify at the largest facilities. Without further innovations in hiring and reductions in training times, it would take an average of at least 4.8 years from tentative hire offer for new hires to reach CPC status in FAA's En Route Centers. For CPCs required at the largest Terminals, the bulk of which rely on transfers from smaller facilities, it would take an average of at least 5.5 years from tentative hire offer for new hires to reach CPC status, primarily because of the average period of 1.9 years that elapses between first certification to CPC level and transfer to a larger facility.

Model Comparisons and Constraints

The ongoing shortage of controllers does not necessarily mean FAA's staffing models are flawed or need to be replaced. The committee's report compares two modeling approaches intended to meet FAA's need to estimate adequate staffing levels for safety and efficiency: the agency's traditional modeling approach and a newer collaborative approach developed with NATCA.

² The National Academies have advised FAA on models to estimate the adequacy of ATC facility staffing in three previous reports published in 1997, 2010, and 2014.

³ Additional elements of the committee's task are addressed in the report.

- The traditional approach has more strengths and fewer weaknesses than the collaborative approach in the committee's view. Significantly, the former can forecast staffing to support future operational demand, while the latter cannot. However, the suite of models that underpin the traditional approach has not been updated to reflect changes in ATC technology and procedures over the past decade or more. Also, the traditional approach may underestimate the time required for training and other duties that controllers perform when they are not on position.
- A key assumption of the collaborative approach—that no other duties can be performed by controllers during slow traffic periods—most likely overestimates the time needed to perform these other duties and thus may overestimate the number of controllers required at individual facilities. An advantage of the collaborative approach is its reliance on a survey of facility managers, which can identify unique operating practices at individual facilities that models cannot represent.

For both the traditional and collaborative approaches, the accuracy of their respective facility staffing targets has not been established. Because of the common use of overtime irrespective of staffing levels, reliance on overtime is not useful for evaluating model accuracy. The committee therefore recommends methods of assessing modeled target accuracy using data the committee did not have access to as well as additional safety metrics described in the report.

Controller Fatigue and Safety

The NASEM 2014 report made many recommendations to FAA, most of which the agency adopted in whole or in part. The most important recommendations addressed the fatigue associated with understaffed facilities that could compromise safety. Fatigue among controllers is affected by facilities' long-time reliance on weekly fatigue-inducing shift schedules that rotate backward against circadian rhythms. Many controllers prefer this type of schedule because it compresses their workweek and gives them at least 80 hours off

between weekly shifts. However, the NASEM 2014 report cautioned against using such shift patterns because they do not allow for sufficient restorative sleep, particularly before the midnight shift. In response to the recommendations of an FAA-convened expert panel's 2024 recommendations, which parallel those of the NASEM 2014 report, FAA mandated longer recovery periods between shifts in FY 2024 (to occur in FY 2025) and initiated the elimination of backward rotating shifts (expected in FY 2026).

Recommendations from the NASEM 2014 report not fully adopted or implemented include

- Research to evaluate the relationships between safety risks and facility staffing levels, the lack of which makes it difficult to isolate the effects of past widespread use of counterclockwise shift schedules on fatigue from the effects of understaffing.
- Management of fatigue inherent in 24/7/365 operations through a robust Fatigue Risk Management System (FRMS), and
- Acquisition and implementation of shift scheduling software that allows for conformance with fatigue rules and principles and efficient shift scheduling of available staff.

REPORT RECOMMENDATIONS TO FAA AND CONGRESS

Based on its findings, discussed further in its report, the committee's main recommendations are listed below.

RESOLVE INADEQUATE FACILITY STAFFING LEVELS AND PERPLEXING SCHEDULING DATA

Even as it improves its models' estimates as recommended below, FAA should make a concerted effort to staff individual facilities based on what its models estimate, particularly those that have an outsized impact on delays in commercial aviation. As part of this process, FAA should use increased incentives to induce more transfers of CPCs from overstaffed to understaffed facilities.

Congress should provide funding for (a) needed hiring to reach modeled staffing standards based on adjustments recommended below and (b) additional incentives for transfers from overstaffed to understaffed facilities.

FAA should determine the causes of the perplexing trends in overtime, regular hours worked, and TOP and rectify them as needed.

UPDATE AND APPLY FAA'S TRADITIONAL MODELING METHODS

FAA should return to the practice of adjusting facility staffing targets estimated with its traditional approach with input from the field for facilities with unique operating conditions by using a structured survey process such as the one used in the collaborative modeling approach.

FAA should reexamine its task load requirements models and whether its estimated facility staffing standards provide adequate time for other duties, arrange for any necessary updates, and establish a system to periodically evaluate the need for future updates.

FAA should use the confidential and other safety indicator data that it collects to validate, to the extent possible, its traditional facility staffing models.

RESPOND TO FATIGUE AND SAFETY CONCERNS WITH RESEARCH, ORGANIZATIONAL CHANGES, AND RECOMMENDED SHIFT SCHEDULING TOOLS

FAA should fully implement previously recommended research on the relationship between staffing levels and safety risks, adjust its facility staffing model parameters accordingly, and implement the recommended robust FRMS and efficient shift scheduling tools. The latter should include rigorous requirements and a Request for Proposals process that includes a build versus buy decision for shift-scheduling software. This process should also include implementation and organizational change management support for greatest likelihood of successful implementation and adoption.

Congress should provide any additional funding required to increase FAA expertise in FRMS and for it to acquire (or develop) and implement the recommended shift scheduling tools.

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FOR MORE INFORMATION

This Consensus Study Report Highlights was prepared by the National Academies' Consensus and Advisory Studies Division based on the report *The Air Traffic Controller Workforce Imperative: Staffing Models and Their Implementation to Ensure Safe and Efficient Airspace Operations* (2025).

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